

BIO - DATA

1. Name : Prof. P. Ramakrishna
2. Age : 77 years (08-07-1948)
3. Qualification : B. Tech (Chem. Engg.)
M. Tech. (Chem. Eng.)
FAFST(Fellow of Association of Food Scientists and Technologists)
4. Specialization : Food Processing Technology
H R D (Teaching & Training)
QA (Testing and Quality control
R&D (Process/Product Development)
Industrial consultancy/Technology Transfer
5. Experience : 16 years at Tata Tea, Limited in R&D (1990-2006)
15 Years at CFTRI in R&D(1975-1990)

12 years in Academic after retirement
11 years at JNTUK in HRD & QA(2011-2022)
3 years at JNTUA in HRD & QA(2018-21)
2 years at AKNU(2023-25)
6. Positions held : Professor of Practice, Adikavi Nannaya University, Rajahmundry(2023-25)
Academic Advisor/Visiting Professor, JNTUK (2021-22, till 16-05-2022)
Program Director, School of Food Technology, JNTUK (2011-2019)
Head, Food Testing Laboratory, JNTUK (2014-2019)
Visiting/Emeritus Professor-Food Technology, JNTUACEK (2018-21)
GM & Head, R&D, Tata Tea (1990-2006)
Scientist In Charge, Process Engineering, CFTRI, Mysore (1975-1990)
7. Contact details: 70-3-36/3, CT Colony
Ramanayya Peta, Kakinada- 533005
07702653444; 09949794222(Mobile)
E-mail:
ramakrishna.p@hotmail.com

8. Major R&D contributions:

Sl.No	Title	Nature of Contribution
I	Development of a fractionation system for Red chillies	A process for benefication of pungent portion was developed
II	Improved process for cleaning, drying and washing of black pepper	A continuous washing and drying system was developed and demonstrated
III	Improvements in the process of manufacture of garlic powder	A continuous method for dehulling and husk separation for the dried cloves was developed, released to 7 parties and at least 2 of them in production
IV	Development of a fixed counter current battery for instant coffee	Pilot conditions for high pressure process to get 46% yield on R/G coffee were standardised
V	Development of a continuous extraction plant for instant Tea	Preferential leaching process was developed to get good flavour in the end product
VI	Process scale-up studies on the production of pectin from apple pomace	A concentration and precipitation process for production of pectin was developed. Conditions for spray drying of pectin are also
VII	Development of a dehulling system for muskmelon, watermelon and other allied fruit seeds	A mechanical dehulling process was developed. Process released to 7 parties. Two units are in production
VIII	Dehydrated flat potato chips/slices	A method for flat slices was developed and demonstrated to about 8 parties
IX	Mechanisation of Makhana processing	Developed a puffing unit and demonstrated
X	Flavoured Tea	Product Developed with 14 different flavours
XI	Decaffeinated Tea	Process Knowhow Developed
XII	Double effect evaporator	Design, Development and testing
XIII	Coffee Bag	Product Released
XIV	Energy Management in Tea Manufacture	Improved thermal efficiency by 25% by Waste Heat Recovery
XV	High Cuppage Tea	Enzymes were used to increase Cuppage and Quality

Sl.No	Title	Nature of Contribution
XVI	Aqueous Tea Aroma	Process Knowhow Developed
XVII	Process controls in Tea Manufacture	Microprocessor based controls were tried for the first time
XVIII	Cold brew Tea	Product Development
XIX	Green Tea	Product Development
XX	RTD	Product Development
XXI	Vitamin Tea	Product Development
XXII	Tea cubes	Product Development
XXIV	Fast Brew Tea	Product Development
XXV	Flavored green Tea	Product Development
XXVI	Improvements in Tea Processing	Use of pectinase as processing aid to improve TF/TR. PFA clearance obtained.
XXVII	Iron filing in Tea	Liaison with PFA to fix limits at 200 ppm

9. Major Consultancy assignments:

i) Fruit and vegetable processing plant at Hazipur, Bihar

As the process knowhow for 50 ton/day Tomato plant was not available, the Bihar Fruits and Vegetable Corporation was planning to import the technology. CFTRI then undertook the responsibility of establishing a totally indigenous unit and designed a double effect evaporator of 1700 kg/hr water evaporation which was fabricated by M/s. L&T Ltd., Bombay under CFTRI supervision. The entire plant was commissioned on turn-key basis using indigenous equipment in 1985.

Capacity	:	50 TPD Tomato, 20 TPD Pineapple, 10 TPD Mango, 5 TPD Lichi, 6000 Bottles per day RTS
Capital	:	2 Crores
Employment potential	:	50 Persons
Value added	:	3 crores per annum
Individual contribution	:	Coordinated the project implementation and commissioning

ii) Fruit processing plant at Nigmoi, Arunachal Pradesh

Arunachal Government retained CFTRI as consultant for setting up of this unit on turnkey basis during 1987. This was the first indigenous Pineapple juice concentration plant using a double effect Plate evaporator in India

Capacity	:	5T per shift of Malta orange or Pineapple
Capital	:	1.5 Crore
Employment potential	:	100 Persons
Value added	:	1 crore per annum
Individual contribution	:	Commissioning of concentration plant

iii) Tomato and malta orange processing plant at Abohar, Punjab

CFTRI revised the report prepared by consultancy cell of Food ministry and established this unit in 1984 for M/s. PAIC. Initially a totally indigenous plant was suggested, but later capacity of the plant was increased and certain equipment like Centitherm evaporator was imported to improve the project viability.

Capacity	:	5T per hour Malta orange and 30 TPD Tomato
Capital	:	1.5 Crore
Employment potential	:	60 Persons
Individual contribution	:	Project appraisal

10. Important Project/Technical Reports:

Sl.No.	Project Report	Client(s)
i	Manufacture of garlic powder	a) Technical Consultancy Organization of Karnataka, Bangalore b) Bihar Pharmaceuticals and Chemicals Development Corporation c) Garlic Corporation, Jaora d) Dehumidified and Hydralyse Processing Pvt. Ltd., Rajkot
ii	Manufacture of green chilly powder	Poonamchand Tiwari, Khandwa
iii	Manufacture of pepper powder	K.S.C.M.F Ltd., Ernakulam
iv	Manufacture of mustard powder	N.R.D.C., New Delhi
v	Manufacture of spices, condiments and masala powders	Technical Consultancy Organisation of Karnataka, Bangalore
vi	Red chillies fractionation	N.R.D.C., New Delhi
vii	Manufacture of biscuits	Calypso Products, Trivandrum
viii	Status and scope of food industry in U.P.	Govt. of U.P.
ix	Food industry in Haryana present status and future scope	Govt. of Haryana
x	Iron filings in Tea	Indian Tea Association. Liaised with PFA to fix the limits at 200ppm

11. Patents:

- i) A preferential leaching process for the preparation of tea infusion with retention of natural flavours

P. Ramakrishna, S. Nagalakshmi, M.S. Ramaswamy,
R. Seshadri, P.K. Ramanathan and C.P. Natarajan

Patent No: DEL/638/1983

- ii) A method for enhanced dewaxing of crude rice bran oil

P. Ramakrishna, B. Manohar, K. Udayasankar and
K.V.L. Venkatesh

Patent No: DEL/976/1988

- iii) A multistage countercurrent Fluid bed drier for drying fermented tea leaves and other similar materials

P. Ramakrishna

Patent No. 889/Cal/1990

- iv) A multistage countercurrent Pneumatic drier for drying fermented tea leaves and other similar materials

P. Ramakrishna

Patent No. 890/Cal/1990

- v) An automatic device for Detoxification of aflatoxin contaminated edible oils by Photolysis
T. Shantha, K. Venkatesh Murthy, S. T. G. Jayaprakashan, K. M. S. Raghavarao and P. Ramakrishna

Patent No. 266/Del/2000

- vi) Method of processing Green Tea leaves to produce Black Tea preparation that can be brewed in cold water

P. Jose David, P. Ramakrishna, A. Bharadwaj and S. Ravi

US Patent 6761918 B2/July 13, 2004

12 Publications:

- i) Chemistry and Technology of melon seeds
M.S. Teotia and P. Ramakrishna
J. Fd. Sci. & Tech., 21(4), 1984, p.332-340
- ii) Melon seeds - Evaluation of physical characteristics
P. Ramakrishna
J. Fd. Sci. & Tech., 23(3), 1986, 1986, p.158-160
- iii) A simple method of manufacture of flat potato chips
P. Ramakrishna and Kamal Prakash
Indian Food Packer 40(6), 1986, p.12-15
- iv) Estimate optimum number of effects for multi effect evaporation
P. Ramakrishna
Chemical Engineering, 94(7), 1987, p.82
- v) Effect of wax content on flow properties of rice bran oil
P. Ramakrishna, K.V.L. Venkatesh, T.C.Poornima and B. Manohar
J.A.O.C.S., 64(6), 1987, p.859-61
- vi) Strength, moisture content and extractability of rice bran pellets
P. Ramakrishna, R. Subramanian, B. Manohar and K.V.L. Venkatesh
J. Fd. Engg., 7(4), 1988, p.281-287
- vii) Some Engineering properties of Pumpkin seeds
M.S. Teotia, P. Ramakrishna, S.K. Berry and Mrs. S. Kaur
J. Fd. Engg., 9(2), 1989, p.153-162
- viii) Densities of Melon seeds, kernels and hulls
M.S. Teotia and P. Ramakrishna
J. Fd. Engg., 9(3), 1989, p.231-236
- ix) Rheological behaviour of clarified Banana juice: Effect of temperature and concentration
Khaleel-E-Khaleel, P. Ramakrishna, A.M. Nanjundaswamy and M.V. Patwardhan
J. Fd. Engg., 10(3), 1989, p.231-240
- x) Effect of Pectin content on Flow properties of mango pulp concentrates
B. Manohar, P. Ramakrishna and R.S. Ramteke
J. Texture Studies, 21, 1990, p. 179-190
- xi) A generalised chart for Breakeven and sensitivity analysis
P. Ramakrishna
Indian Food Industry, 9(4), 1990, p.24-27
- xii) Some physical Properties of Tamarind juice concentrates
B. Manohar, P. Ramakrishna and K. Udayasankar
J. Fd. Engg., 13(4), 1991, p.241-58
- xiii) How often to clean an Evaporator
P. Ramakrishna
Chemical Engineering 98(7), 1991, p.143-146

- xiv) Energy consumption for Tea processing in High Range
P. Ramakrishna, P. Joseph George, M. Ramar and R.D.Lai
PEN News Letter, 2(2), 1991, p.4-9
- xv) Ratio analysis in Tea Manufacture
P. Ramakrishna and R.D. Lai
The Planters' Chronicle, 87(2), 1992, p.71-77
- xvi) Wither Chart
P. Ramakrishna and R.D. Lai
The Planters' Chronicle 87(6), 1992, p.203-5
- xvii) Out-turn Charts
P. Ramakrishna and R.D. Lai
Sri Lanka Journal of Tea Science, 60(2), 1991, p.9-17
- xviii) Optimum length of pruning cycles
P. Ramakrishna
The Planters' Chronicle, 91(1), 1996, p.7-12
- xix) Heat Pipe Heat Exchanger for Waste Heat Recovery
P. Ramakrishna, P.D. Suresh Kumar, P. Joseph George and Koshy Chacko
Sri Lanka Journal of Tea Science , 64(1/2), 1996, 35-42
- xx) Scheduling of fermenting drums for optimum capacity utilisation
P. Ramakrishna
The Planters' Chronicle 90(11), 1995, p.557-9
- xxi) Factors contributing to blackness of tea
P. Ramakrishna and P. Jose David
The Planters' Chronicle 91(3), 1996, p.109-13
- xxii) Chart for determining plucker requirement
P. Ramakrishna
Sri Lanka Journal of Tea Scienc
- xxiii) Importance of SWIR BAND in discrimination of TEA plantations-A case study
K. Gupta, K. Ganesha Raj, M. G. Candrasekar, P. Haridas and P. Ramakrishna
- xxiv) Energy for Tea Manufacture - A Review
P. Ramakrishna
Proceedings of the international workshop on Global advances in Tea Research , 1997.
- xxv) A rational approach for Nitrogen application in Tea
P. Ramakrishna
The Planters' Chronicle, 1997

13. Papers presented at symposia/seminars:

- i) Improved cleaning, washing and drying of pepper
P. Ramakrishna and P.K. Ramanathan Symp. on Pepper and Pepper products organised in connection with International Pepper Community meeting, Mysore, 18-22 May 1981
- ii) Rheological properties of Banana Juice concentrate
Khaleel-E-Khaleel, P. Ramakrishna, A.M. Nanjundaswamy, M.V. Patwardhan
International Food Convention, Mysore, 18-23 Feb. 1988
- iii) PVC Solar collectors for hot air generation
K. Girish and P. Ramakrishna National Workshop on Solar Energy, Udaipur, 7-8 Apr. 89
- iv) Energy consumption for Tea processing in High Range
P. Ramakrishna, P. Joseph George, M. Ramar and R.D.Lai National seminar on renewable energy in plantation industry, Madurai, 1-3 March 1991
- v) Scope of cogeneration in Tea processing
P. Joseph George, V.V.N. Kishore and P. Ramakrishna International Workshop on Energy Perspectives in Plantation Industry, Coonoor, 10-12 Feb. 1993
- vi) Energy consumption for Tea processing in India-Status and Alternatives
P. Ramakrishna, P. Joseph George, M. Ramar and Gautam Dev International Workshop Energy Perspectives in Plantation Industry, Coonoor, 10-12 Feb. 1993
- vii) Energy Plantation on Waste Land
- A model Man-made Forest developed by Tata Tea
P. Rajasekaran, P. Mohan Kumar, P. Haridas and P. Ramakrishna International Workshop on Global warming issues in Asia, Bangkok, Thailand, 8-10 Sep. 1993
- viii) Extraction, purification and assessment of polyphenols in green and black teas
Kamal K. Biswas, Satyahari Dev, P. Jose David, P. Ramakrishna and B.C. Bhattacharya
CHEMCON - 1994, IIT, Kharagpur
- ix) Emerging trends in Agro-processing and R&D efforts at Tata Tea
P. Ramakrishna
AFST(I) Annual Conference, Sep. '95, Mysore
- x) Energy perspectives for Tea manufacture in Assam
P. Ramakrishna
Business Meet by TERI and ABITA, May '95, Dibrugarh
- xi) Studies on Irrigation in Tea in High Ranges
P. Haridas and P. Ramakrishna
Seminar by UPASI, Jan. '96, Coimbatore

- xii Irrigation potentials of High Ranges
P. Ramakrishna
Inhouse Seminar by Tata Tea Ltd., Munnar, 1995
- xiii) Computerised Process Controls in Tea Manufacture
Koshy Chacko and P.Ramakrishna International workshop on global advances in Tea science New Delhi, 1996
- xiv) Scope of Electronic Process Controls in Tea Manufacture
P.Ramakrishna
UPASI Conference, Conoor, 1999

Articles published in in-house Magazines

- i) Programme Plucking
P. Ramakrishna
Tata Tea Seithigal, 2(5), 1996, p.6-7
- ii) Coorg Coffee Bag - The inside story
P. Ramakrishna, Harish Bijoor and Rabin Lai
Tatean, v(i), 1996, p.6-7

14. Visits Abroad:

Country	Place(s) visited	Subject(s)
USA	SNAPLE, New york Tata Tea Inc., Florida	RTD Soluble Tea
KENYA	African Highlands Produce Co. Ltd.	Tea cultivation & Manufacture
	Tea Research Foundation of Kenya	Tea Research
	Finlay Flowers, Kericho	Cut flowers
HOLLAND	Ofman Verbeek International B.V.	Cut flowers, TC plants and their market
	Verenigde Bloemenveilingen Aalsmeer (VBA)	Auction of cut flowers
SWITZERLAND	Nova Swiss, Nova Werke AG SITEC, Sitec Sieber Engg. AG	Super Critical Fluid Extraction Super Critical Fluid Extraction
UK	Lyons Tetley Ltd James Finlay & Co London Chamber of Commerce, London	Tea Packaging Tea tasting Tea Auction
FRANCE	Paris	Tea Marketing
SINGAPORE	Plantek International (Pte) Ltd.	Tissue Culture
THAILAND	Kasetsart Universty, Bangkok Asian Institute of Technology, Bangkok Bangkok Flowers Centre Co. Ltd. FAO, Bangkok	Food Technology Energy management Tissue culture Wood energy
BELGIUM	Co-packers of Tetley	RTD factory
BRUSSELS	Tea Board delegation	Pesticide Residues in Tea
GERMANY	Co-packers of Tetley	Instant Tea factory
SRILANKA	Tea Research Institute Watawala Plantations	Tea Research Tea Estates/Processing

15. Teaching and Training:

- i) Course lecturer for 2-year M.Sc (Food Tech.) course conducted by CFTRI, Mysore during 1977-1981 (4 batches)
- ii) Course coordinator for M.Sc (Food Tech.) during 1985-90 (5 batches). Revised the syllabus and introduced SI units. More emphasis was given to solving problems in food engineering
- iii) Visiting Faculty in Food Technology, SV University College of arts & Science, 2006
- iv) Initiated basic work on Engineering properties of Food at CFTRI
- v) Designed and Conducted a short term course on “Evaporators for fruit juice concentration” during December 1988
- vi) Prepared a manual on Guidelines on Tea Manufacture for the factory staff at Tata Tea
- vii) Conducted courses on Tea Manufacture at Tata Tea. Set up Tea Testing Laboratory
- viii) Set up a pilot plant for Tea Manufacture which is first of its kind, at Tata Tea, Munnar
- ix) Initiated work on New Product Development at Tata Tea. Set up product development Labs. and Pilot Plants at Tata Tea, Bangalore
- x) Guidance to students:

Name	Topic
Mr. Min Young Kyoo	Dissertation on “Design of Spray Dryers” for M.Sc - 1977
Mr.Khaleel-E-Khaleel	“Engineering properties of Banana juice concentrate” as a part of Ph.D - 1989
Mr. Jiji Austin	“Plant layout for CTC Tea Factory” for B.E. -1991
Mr. Kamal Biswas	“Extraction, purification and quantification of metabolites from invitro and invivo tissues of Camellia sinensis, Centrella asiatica and Acorus calamus” for M.Tech - 1994
Ms. R. Lakshmi	Development of “Software for Fertiliser application for Tea” for B.Sc -1996

- xi) Established School of Food Technology at JNTUK. Designed curriculum and syllabus for M. Tech (Food Processing Technology) course and it is one of the best such courses in India with state of art facilities. Coordinated the course during 2011-2019. Started B. Tech in Food Engineering in 2020-21, first such course in AP.

A Food Testing Laboratory has been set up at School of Food Technology, JNTUK, Kakinada to assist the regional Food Industry in quality control, with grant of 156 lakhs from MoFPI. First such lab. in AP with NABL accreditation and FSSAI recognition. Veg. oils and other materials imported by Kakinada port are analyzed regularly. Lab is up graded with grant in aid of 339 lakhs from APFPS. Lab is generating revenues of Rs 2 lakhs/ month.
- xii) Started B. Tech (Food Technology) course at JNTUA College of Engineering, Kalikiri. in 2018-19 Laboratories set up. Basic Food Testing Laboratory was also setup from grant Rs. 6.39 crores from MoFPI
- xiii) Started M. Sc in Food Science and Technology course in 2023-24 at Adikavi Nannaya University. Special feature of the course is project work in IV sem will be carried out at CFTRI, Mysore. Rs. 3 Crores sanctioned under PM USHA.for State of art Laboratories.

16. Member of Research and Academic Committees:

Member of Research Advisory Council for Institute of Himalayan Bio-resource Technology, Palampur (1998-2001)

Member of Technical Committee for Tea - Bureau of Indian Standards (1995-2005)

Member (2001) and chairman (2002) of Technical Committee - Indian Tea Association

Member of engineering committee-TRA(1998-2002)

Member of editorial board for the Journal of Food Science and Technology (1989)

Member of editorial board for Indian Food Industry (1989)

Life member of Indian Institute of Chemical Engineers

Life member of Association of Food Scientists and Technologists (India)

Life member of Indian Society for Plantation Crops

President, Association of Food Scientists and Technologists, India (2005)

Member of Scientific panel on Water & Beverages (including alcoholic beverages), FSSAI (2015-19)

Member of Scientific panel on Spices & culinary herbs, FSSAI (2018-19)

Member of Scientific panel on Water & Beverages, FSSAI (2020-22)

Member of Task Force on Food Processing in AP (2018)

Reviewer for Journal of Food Science and Technology, Indian Food Industry and Journal of Food Engineering (UK).

Research council member & Consortium Coordinator- Quality and post-harvest Technology, National Tea Research Foundation, Tea Board (2018-21)

Member/Chairmn of BOS- Food Technology, JNTUK

Chairman of BOS- Food Technology, JNTUA

Member of Advisory committee for B. Tech (Food Technology), JNTUA

17. Awards

- i) AFST (I) Conferred 'Fellow of Association of Food Scientists and Technologists' (2005) for his outstanding contributions in the area of Food Science and Technology.
- ii) Andhra University Nominated me Honorary Professor (Emeritus) in the department of Food, Nutrition & Dietetic by (2022-24)